

NATIONAL EXAM OF HIGHER NATIONAL DIPLOMA– 2023 SESSION

Specialty/Option: *Software Engineering*

Paper: *Practice of Computer*

Duration: 5 hours

Credit Value: 08

SECTION A: ALGORITHM AND PROGRAMMING

/50 marks

I- C programming implementing a Simple Temperature Conversion Program 25 MARKS

You have been tasked with implementing a simple temperature conversion program using C programming. The program should prompt the user to enter a temperature in Fahrenheit and convert it to Celsius. The program should output the result to the console.

1. What is a flowchart in programming? List its components (5 marks)
2. Draw the flowchart corresponding to the scenario (6 marks)
3. To implement this program, you will need to:
 - a. Implement user input. Your program should prompt the user for the temperature in Fahrenheit and read in the user's input. (3 marks)
 - b. Implement temperature conversion. Your program should use the formula $(F - 32) * 5/9$ to convert the temperature from Fahrenheit to Celsius. (3 marks)
 - c. Implement error handling. Your program should handle errors such as invalid input (e.g., non-numeric values for the temperature). (5 marks)
 - d. Output the result. Your program should output the converted temperature to the console. (3 marks)

Your task is to implement a working prototype of this program, including the four components listed above. During the practical exam, you will be evaluated based on your ability to write clean, efficient, and well-documented C code that implements the temperature conversion logic correctly. You will also be evaluated on your ability to handle errors and debug your code to ensure that it works as expected.

II- Object oriented programming

25 MARKS

Implementing a Simple Library Management System using Object-Oriented Programming

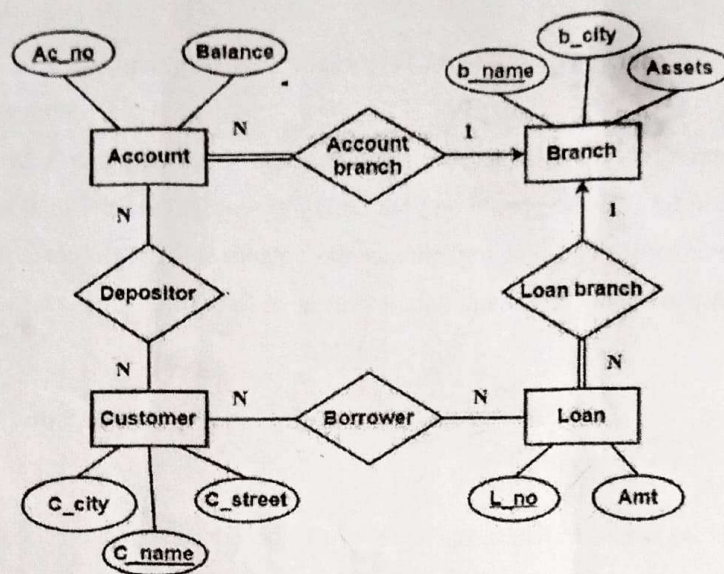
You have been tasked with implementing a simple library management system using object-oriented programming principles. The program should allow the user to add new books, remove books, and display a list of all books in the library. Each book should have a unique identifier, title, author, and number of copies available.

1. What is a class? Differentiate it from an object (4 marks)
2. To implement this program, you will need to:
 - a. Define a class for the book. Your class should have member variables for the book identifier, title, author, and number of copies available. (4 marks)
 - b. Implement constructors and getters/setters. Your class should have a constructor to initialize the member variables and getters/setters to access and modify the member variables. (5 marks)
 - c. Implement methods to add and remove books. Your program should allow the user to add new books to the library and remove existing books from the library. (4 marks)
 - d. Implement a method to display all books. Your program should allow the user to display a list of all books in the library. (4 marks)
 - e. Implement error handling. Your program should handle errors such as attempts to remove non-existent books. (4 marks)

Your task is to implement a working prototype of this program, including the five components listed above. During the practical exam, you will be evaluated based on your ability to write clean, efficient, and well-documented object-oriented code that implements the library management system logic correctly. You will also be evaluated on your ability to handle errors and debug your code to ensure that it works as expected.

SECTION B: DATABASE DEVELOPMENT AND ADMINISTRATION (MYSQL) /20 marks

To design an information system of a bank we have the following diagram:



- 1- What is this representing? Describe it with all his components (4 marks)
- 2- Give the equivalent relational model of this figure (5 marks)
- 3- Using appropriate software to create a database entitled BANK and create all the tables and populate each with 3 data's (8 marks)
- 4- Write a query to select all customer having 3 accounts and which have borrow 1500000 FCFA in your database and you screenshot it and save with the name HNDWE2022 in a folder HND2023 (3 marks)

SECTION C: WEB DESIGN

/15 marks

1. In others to design and executes a website, what do we need? (1.5 marks)
2. Differentiate between static and dynamic website (1.5 marks)
3. Implementing a Simple Contact Form using HTML, CSS, and JavaScript

You have been tasked with implementing a simple contact form using HTML, CSS, and JavaScript. The contact form should allow the user to input their name, email address, and message, and submit the form.

To implement this program, you will need to:

- a. Design the user interface. Your contact form should have a clean and user-friendly interface that displays the input fields, allows the user to input their name, email address, and message, and provides a submit button. (3 marks)
- b. Implement the contact form logic using JavaScript. Your contact form should use JavaScript to validate the input fields and send the form data to a server. (3 marks)

- c. Implement error handling. Your program should handle errors such as attempts to submit empty fields or invalid email addresses. (3 marks)
- d. Style the contact form using CSS. Your program should be visually appealing and use CSS to style the UI. (3 marks)

Your task is to implement a working prototype of this program, including the four components listed above. During the practical exam, you will be evaluated based on your ability to write clean, efficient, and well-documented HTML, CSS, and JavaScript code that implements the contact form logic correctly. You will also be evaluated on your ability to handle errors and debug your code to ensure that it works as expected.

SECTION D: NETWORKING

/15 marks

Scenario:

A small business called "Cafe Connect" has three computers and a printer that need to be connected to a network. The business owner wants to be able to share files between computers and access the printer from any computer on the network.

Tasks:

1. List the various devices which can be used to setup the network
2. Design a network diagram for Cafe Connect that includes the following devices: switch and three computers.
3. Configure the devices in the network diagram to provide the following functions:
 - Assign IP addresses to all devices in the network.
 - Configure the switch to provide connectivity between all devices in the network.
 - Share files between the three computers.
 - Enable the printer to be accessible from any computer on the network.
3. Test the network to ensure it is functioning properly.
4. Document the network diagram and device configurations.

Time allocation:

- Network design: 2 marks
- Device configuration: 5 marks
- Network testing: 2 marks
- Documentation: 1 mark